YUVANSARAVANAN R

Computer Science Graduate | Java & Python Developer | Machine Learning Engineer
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PROFESSIONAL SUMMARY

Computer Science graduate (2025) with hands-on experience in Java, Python, Machine Learning, and full-stack web development through academic and personal projects. Skilled in building REST APIs with Spring Boot, developing CNN-based computer vision models, and applying reinforcement learning to autonomous systems. Strong foundation in data structures, databases, model evaluation, and the software development life cycle, with a proven ability to learn new technologies quickly and deliver working, deployable solutions.

CORE SKILLS

Programming Languages: Java, Python, SQL

Artificial Intelligence & Machine Learning: Machine Learning, Deep Learning, Convolutional Neural Networks (CNN), Reinforcement Learning, Computer Vision, Image Classification & Recognition, Model Training & Evaluation, Model Optimization, Data Preprocessing

AI/ML Frameworks & Libraries: TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib

Web Technologies: HTML, CSS, JavaScript, REST APIs, Spring Boot

Database: MySQL

Cloud, Tools & Platforms: AWS (Basics), Git, GitHub, CARLA Simulator, MS Excel, MS Office

Data Analytics: Data Analytics using Python, Exploratory Data Analysis, Data Visualization

Soft Skills: Problem Solving, Team Collaboration, Time Management, Adaptability, Communication, Leadership, Public Speaking, Event Coordination, Decision Making, Critical Thinking, Analytical Thinking, Mentoring

PROJECTS

Autonomous Navigation System for Self-Driving Cars Using Reinforcement Learning

Technologies: *Python, Reinforcement Learning, CARLA Simulator, Deep Learning*

- Researched reinforcement learning techniques for autonomous driving by studying existing methodologies and academic literature.
- Collaborated in a team to design self-driving car navigation logic using the CARLA simulator; findings were published in an international conference paper.

Library Management System

Technologies: *Java, Spring Boot, MySQL, REST APIs*

- Built a web-based application to manage books and members, with features for adding, updating, issuing, and returning books.
- Integrated MySQL with Spring Boot for persistent data storage; built RESTful APIs with input validation and exception handling.

Image Recognition System

Technologies: *Python, TensorFlow, Keras, OpenCV, CNN*

- Developed a deep learning-based image classification system using a Convolutional Neural Network (CNN).
- Performed image preprocessing and data augmentation, trained and validated the model, and evaluated and optimized accuracy using Python ML libraries.

Automatic Solar Grass Cutter Using Microcontroller

Technologies: *Embedded Systems, Microcontrollers, Solar Panel, DC Motors, Sensors*

- Designed and fabricated an autonomous, solar-powered lawn mower to reduce fuel dependency and carbon footprint.
- Programmed a microcontroller to manage power distribution and motor speed, and implemented sensor-based obstacle detection for safe navigation.

EDUCATION

Bachelor of Engineering (B.E.), Computer Science and Engineering *2021 - 2025*
KPR Institute of Engineering and Technology | GPA: 8.47 / 10.0

Higher Secondary Certificate (HSC) *2020 - 2021*
Kongu Matriculation Higher Secondary School | Percentage: 92%

Secondary School Leaving Certificate (SSLC) *2018 - 2019*
Kongu Matriculation Higher Secondary School | Percentage: 88.8%

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Vice President, Quiz Club *2022 - 2023*
- Led club activities and coordinated quiz events, overseeing planning and team participation for the academic year.

CERTIFICATIONS

- IBM: Python Programming
- HackerRank: Python and Java Coding (Badges)
- NPTEL: Data Analytics using Python
- NPTEL: Simple Mobile App Development for a Small Startup Company

LANGUAGES

English (Fluent) | Tamil (Native)